

□ - [] **Course Syllabus: *Segnali Elettrofisiologici* Master's Degree in Bioengineering | UNIMORE**

- **Academic Year:** 2026/27
- **Course code:** BIOING_20
- **Course info:** year 2, winter semester, 6 credits
- **Lecturer:** Prof. Eng. Michele GIUGLIANO, PhD
- **Format:** In-person lectures
- **Attendance:** At least 80%

Syllabus: It is an outline of the topics taught in the course. It also contains additional information, designed to 1) preemptively answer students' questions, 2) list policy and mutual expectations and 3) the consequences of failing to meet them. It does not replace the official information about the course, published online at <https://unimore.coursecatalogue.cineca.it>, but extends it.

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1. Learning Objectives

tudents will be exposed to a selection of key topics in bioengineering, bringing them into direct contact with industrial, business, and academic environments. Upon completion of the course, students should have developed advanced knowledge and analytical skills in applying biomedical engineering principles to the development of innovative medical solutions. The course aims to develop the ability to critically evaluate and apply cutting-edge methodologies in the design and optimization of medical technologies, preparing future bioengineers to address complex challenges in the field of innovative and translational medicine.

2. Prerequisite knowledge

Curricular requirements declared: By enrolling in this Master's Degree program, students have formally declared they satisfied the UNIMORE curric-

ular requirements. For the purposes of this specific course, this translates to an expected, solid foundation in A. Mathematics & Statistics, B. Physics & Biophysics, C. Chemistry & Physical Chemistry.

Students with lack of prerequisite knowledge: Given the interdisciplinary nature of Bioengineering, students may occasionally discover significant gaps in their foundation knowledge throughout the classes. If a student identifies such gaps, **they are expected to contact the Lecturer as early as possible in the semester.** The Lecturer is fully committed to supporting student's extra learning, but **assistance can only be provided if these challenges are communicated proactively.** If the Lecturer is not made aware of specific deficiencies in advance, it will be impossible to offer the necessary support or tailored guidance later in the course.

Core Class Content

The course is designed as a series of guest lectures and interactive discussions led by distinguished national and international experts from academia, industry, and clinical practice. Each session will typically feature an extended Q&A with the speaker, offering a comprehensive perspective on cutting-edge advancements and the translational journey of biomedical innovation, encompassing technical, academic, and entrepreneurial aspects of a career in bioengineering.

I. Advanced Bio-Interfaces, Diagnostics, and Sensory Restoration (2 CFU): This macro-topic explores the latest advancements in medical diagnostics, real-time physiological monitoring, and technologies designed to interface directly with biological systems or restore sensory functions. Specific areas and insights from guest lecturers will include: ◦ Neuroengineering and Neuroprosthetics: Focusing on advanced interfaces and therapeutic applications. ◦ Cochlear Implants and Nerve Interfacing: Technologies for auditory and neural stimulation and their clinical applications. ◦ Ophthalmology, Visual Neuroscience, and Retinal Prosthetics: Exploring electro-cellular interfaces and medical technologies for vision restoration. ◦ Advanced Biosensing and Monitoring Systems: Including the development and impact of Continuous Glucose Monitoring (CGM) systems for diabetes management, and innovations in biosignals monitoring and vital signs tracking.

II. Innovative Bioengineering Therapies and Translational Medicine (2 CFU): This section examines emerging therapeutic strategies, advanced biological models for research, and the critical journey of medical innovation from scientific discovery to patient care. Key discussions and guest perspectives will cover: ◦ Organ-on-a-chip Technologies: Their application in disease modelling, drug discovery, and personalized medicine. ◦ Therapeutic Strategies and CNS Gene Therapy: Including a practical “reality check” on “Bringing medical innovation to patients,”. ◦ Augmented Reality for Surgery: The application of immersive technologies to enhance surgical precision and training.

III. Bioengineering Innovation, Entrepreneurship, and Regulatory Landscape (2 CFU): This macro-topic delves into the entrepreneurial ecosystem within MedTech, the development process of medical devices, and the essential regulatory and commercialisation pathways. Discussions will cover:

- **Bioengineering Entrepreneurship:** Case studies and practical insights into starting and growing companies in the biomedical field.
- **Medical Device Development and Regulatory Pathways:** The lifecycle of medical devices from concept to market access.
- **Bringing Medical Innovation to Patients:** Industry Perspectives: Practical challenges and strategies for clinical translation, intellectual property, and market realities.
- **Startup Innovation in Biosignals and Vitals Tracking:** The entrepreneurial journey in developing and commercialising monitoring technologies.

4. Study Materials, Textbooks, Resources

- **Primary References:** Lectures overheads and reference material, made available by each speaker. All the efforts will be made to insure these material is made available to the students before each class but it cannot be guaranteed.
 - **Readings:** Biomedical Engineering Fundamental, Bronzino, CRC Press
 - **Online Resources and Study Material:** The course has a a) GitHub repository and b) a dedicated website for distributing educational materials and supporting students:
- 1) The **Course Website** (<https://mgiugliano.github.io/colloquia/>) provides a overview of the course, including the updated class schedule, and a list of speakers. In the “Resources” section, links to video tutorials, scientific literature, additional readings, mathematical prerequisites are provided.
 - 2) The **Teams** group serves as a central archive where lecture slides, and reading materials are stored. Students must refer to this material to support their study and dive deeper into the technical and practical aspects of the module.
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5. Grading Procedures

- **Grading for this module:** The assessment of student performance in *this module* is designed to evaluate both continuous engagement with the material and ultimate mastery of the entire curriculum, structured around three distinct components:
 - 1) a **written test** (2.5h, no books, no notes, questionnaire closed-answers, exercises) - (max **30 points**)

2) an **optional** oral exam, as a 20-minute power point presentation on a topic chosen by the Lecturer on a case-by-case basis, necessary to achieve **30 e lode (cum laude)**. **Warning:** This is a comprehensive evaluation, where poor performance may possibly *decrease* the “starting” grade. The Lecturer assigns a topic and one or more scientific papers, that the student must present orally and thoroughly as a “chalk talk” (without any Power Point slide and only with a blackboard).

- **Important Dates:** Exams sessions date (in January, February, June, July, September) are published on esse3.unimore.it. *Note: Students must register at least 10 days in advance for any exam session. Given that this module grants 6 CFUs, students should consider prioritizing this exam over those with a lower number of CFUs.*
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6. Logistics and Policies

Each session follows a similar structure to maximize learning and interaction:

8:30 AM - 9:15 AM: Introduction. The course coordinator will briefly introduce the speaker and their area of expertise.

9:15 AM - 10:15 AM: Formal Talk. A 60-minute presentation where the speaker will delve into their current research, entrepreneurial activities, or primary area of expertise.

10:15 AM - 10:30 AM: Question & Answer Session. A dedicated time for students to ask questions related to the formal talk. Students are encouraged to prepare questions beforehand to “break the ice”.

10:30 AM - 10:40 AM: Short Break. A brief break for refreshments and informal networking.

10:40 AM - 11:10 AM: Career Trajectory Discussion. A 30-minute informal segment where the guest shares their personal and professional journey, offering insights into key decisions, challenges, and milestones. Following this, the session transitions to the hands-on component:

11:10 AM - 12:35 PM: Interactive Hands-on Session. Students will form small groups (3-5 people/group) to collaboratively brainstorm and work on a practical problem, case study, or dataset provided by the speaker. The speaker will remain available to interact with the students, offering guidance and insights. Groups will be shuffled every week, offering every time the opportunity to meet and collaborate with other students (even with different style of interaction, level or proficiency, communication skills).

12:35 PM - 1:30 PM: Group Presentations & Wrap-up. Each group will present a brief summary of their findings, design, (or failure) followed by a

concluding remark and brief discussion with the guest.

- **Videotaped lectures:** As the Master's Degree program in Bioengineering is **not** an "online degree", live lectures are neither systematically recorded nor streamed, unless noted otherwise. No provision for hybrid or online attendance is offered.
- Working students, unable to attend the classes, or those dealing with severe and prolonged illness, should take contact with the Lecturer in due time and with great advance to agree on alternative course material access and study for the exam.
- **Questions:** Students are encouraged to actively ask questions and vigorously participate intellectually **during the class**. As they are early in their training career, students are definitely allowed to ask *stupid* questions. Overly shy students may approach the speaker before the start of a class or during the breaks for quick questions, being aware they are preventing their peers to benefit from the "question and answer". The Lecturer is passionate about transferring his knowledge and fully committed to answer all questions at any time, including "Office Hours" (see Section 8).
- **Attendance Policy:** At least 80% attendance to the classes is expected. Being this module strictly dependent on attendance for class participation and interactive activities, attendance tracking is fundamental.
- Attendance is tracked by the Lecturer for each class via the UNIMORE mobile app (<https://www.unimore.it/it/servizi/unimore-app>). Students **must** install this app on their mobile devices and use it to record their presence when they are in class. The Lecturer provides a unique code at the start of each class (e.g., 9PSTK) to be logged on the UNIMORE mobile app. Students exceptionally arriving late and thus *missing* the unique code communication must immediately alert the Lecturer.
- **Device Policy and Classroom Engagement:** While students will need their smartphones briefly to record attendance via the UNIMORE app, extraneous use of laptops, tablets, and smartphones is strongly discouraged during class to minimize distractions and promote active engagement. The Lecturer does not normally request that students leave their bags and devices at the class entrance, unless for the written exam. Devices inherently bring hundreds of other distractions that are always just one tap or one notification away.
- Students are requested to seriously commit to active participation in each class, demonstrating an investment of attention and focus that matches the Lecturer's own investment of energy, effort, and care in delivering the lecture.
- **On Taking Notes:** Taking handwritten notes is strongly encouraged, as it forces real-time synthesis of the material. Relying solely on the Lecturer's overheads is insufficient for passing the exam.

- **Academic Honesty** Academic dishonesty, plagiarism, and cheating are serious violations of university trust and will result in immediate administrative consequences
 - The Lecturer can also be contacted at any times for information on stages, internships, and master thesis projects.
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7. Teaching Style, Language, and Philosophy

- **Format:** The teaching activity consists of frontal lectures with guest speakers, where key and timely aspects of Bioengineering are introduced and discussed.
 - **Language:** The guest speakers are free to **choose in which language they will deliver their class: English or Italian.**
 - Lecturing in English transforms the classroom into a realistic professional microcosm, offering students a protected, low-risk environment to naturally absorb (and practice) technical terminology before entering the competitive high-tech industry or international research sectors. The Lecturer's proposal to use spoken English during classes is an offer of a strategic investment, designed to bridge the gap between *local* education and the *global* job market.
 - **Language used by students:** Regardless of the chosen lecture language, students can ask questions or seek clarifications speaking either English or Italian during class or office hours. They similarly retain full freedom to choose either Italian or English for their (optional) oral exam.
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8. Contact and Communications

- **Contact:** Students can reach the Lecturer through email (michele.giugliano@unimore.it) or preferably instant messaging (Teams) via

<https://teams.microsoft.com/l/chat/0/0?users=michele.giugliano@unimore.it>

- **Office hours:** The Lecturer provides explanations on the topics covered in the course **only in person, and upon appointment**, and not via Teams or email.
- Students should not hesitate to contact the Lecturer for explanations on any aspect of the course, or to request additional study material and pointers to the literature.
- Students are invited to use Teams to post their questions publicly, so that the entire class may try to answer them, with all benefiting from each interaction.

- **Teams App:** The only official channel used by the Lecturer to communicate to the students is the dedicated **Microsoft Teams group**. Teams is where announcements, posts, information, links, papers, and last-minute or urgent alerts are communicated to the students for their own benefit, interest, as well as for the administrative, logistic, and pedagogical aspects of the course.
- Students enrolled in this course should have been automatically added to the Teams group members by UNIMORE IT services. Students without a UNIMORE email account must **urgently** request the Lecturer, in written (email), to be added to Teams (i.e., *manually*).
- Students should install the Teams app on their device(s) and enable notifications.
- Teams is **not** meant as a unidirectional or passive communication channel. It serves as a **forum** where, **as a community**, students interact with the Lecturer and among themselves by asking/answering each others questions and exchanging information.
- **Student Responsibilities:** Any last-minute communication, sudden schedule changes, or emergency announcements regarding classes and exams will be posted by the Lecturer **exclusively** on Teams. It is entirely students' responsibility to stay informed.
- Missing an urgent update due to unchecked notifications cannot generally be accommodated, except in cases of documented *force majeure*.
- Throughout their **entire** academic tenure as students at UNIMORE, individuals **must** check their Teams app and their unimore.it email account for new posts and messages **every working day, at least once per day**.